

Sustainable Smart City Assistant Using IBM Granite LLM

A Project Report

submitted in partial fulfillment of the requirements

of

Generative AI with IBM Cloud

Internship

with

SMARTBRIDGE in collaboration **with APSCHE**

by

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1. Introduction

1.1 Project Overview

The Sustainable Smart City Assistant is an innovative web application designed to empower urban planners, environmental scientists, and local government officials with advanced, data-driven tools for sustainable city management. It leverages cutting-edge AI via IBM Granite LLM, combining KPI analysis, time-series forecasting, anomaly detection, and AI chat assistance through a seamless Streamlit-FastAPI framework.

1.2 Purpose

To streamline sustainability efforts by enabling real-time KPI analysis, prediction, and eco-guided decision-making, enhancing efficiency and resilience in urban planning.

2. Ideation Phase

2.1 Problem Statement

Urban sustainability is challenged by fragmented systems, poor data usability, and lack of AI integration. This tool bridges the gap between data analytics and decision-making.

2.2 Empathy Map Canvas

Stakeholders: Urban Planners, Scientists, Officials

Say: "We need fast, insightful analysis."

Think: "Smart tools should simplify decision-making."

Do: Upload data, use chat assistant, generate reports.

Feel: Confused by complex data, desire clarity.

2.3 Brainstorming

- KPI uploads in CSV
- Time-series forecasts
- Chat assistant for queries
- AI news summarization (SerpAPI)
- PDF report generation with eco-friendly tips

3. Requirement Analysis

3.1 Customer Journey Map

Awareness: Discovered via GitHub/events

Engagement: Uploads data, interacts with UI

Decision: Uses reports to guide policy

Retention: Regular insights, updates

3.2 Solution Requirement

Functional:

- CSV upload, chatbot, forecasting, anomaly detection
- PDF export, AI summaries

Non-Functional:

- FastAPI backend, <2s responses, Streamlit UI

Constraints:

- No authentication, CSV file storage

3.3 Technology Stack

Frontend: Streamlit, matplotlib, seaborn

Backend: FastAPI, Python 3.11, statsmodels, SerpAPI, IBM Granite LLM

Storage: CSV, optional SQLite/MongoDB

APIs: Watsonx, SerpAPI, optional Pinecone, Google Maps

4. Project Design

4.1 Problem Solution Fit

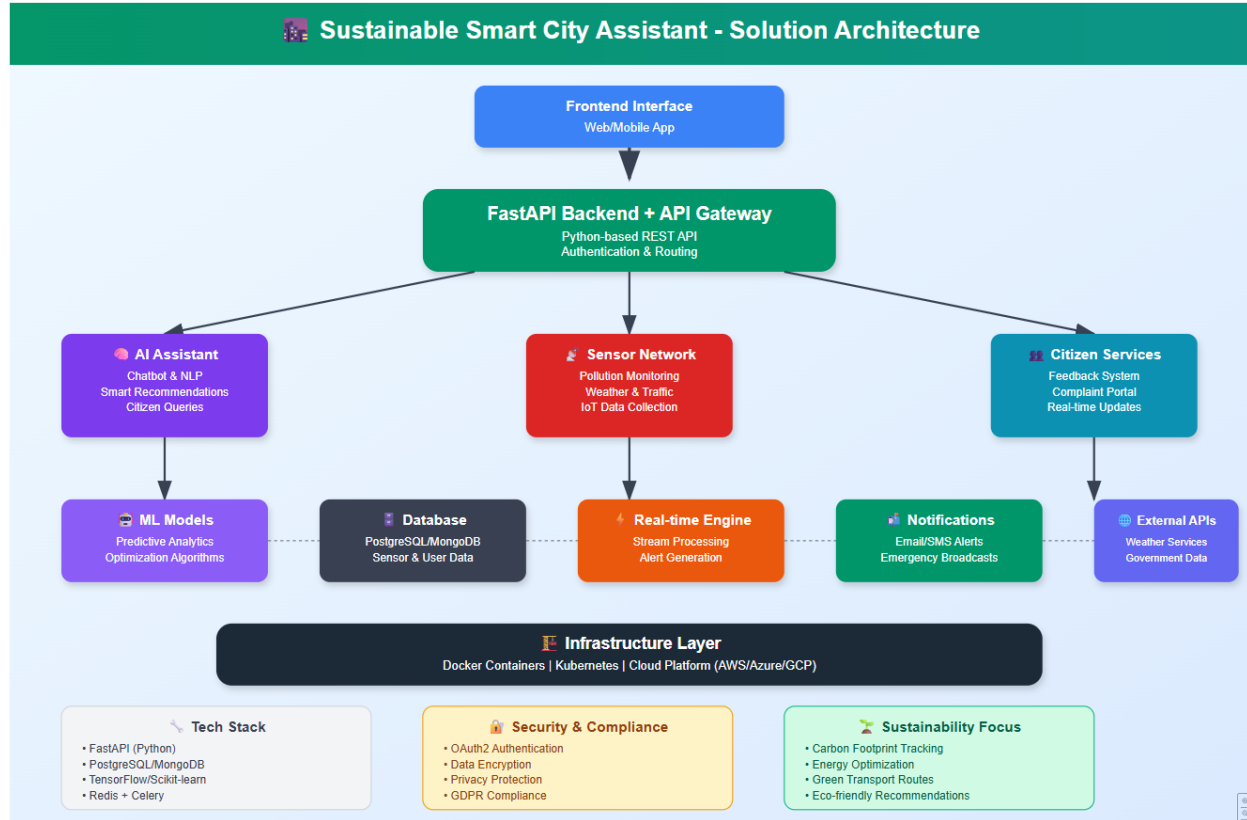
It provides a one-stop, AI-enabled interface to guide eco-conscious city development through accurate data insights.

4.2 Proposed Solution

- Upload city KPI data
- Forecast trends and detect anomalies
- Chatbot interaction with IBM Granite
- Generate PDF reports with sustainability recommendations

4.3 Solution Architecture

See the architecture diagram



5. Project Planning & Scheduling

5.1 Project Planning

Phase 1: Chat assistant integration

Phase 2: CSV upload

Phase 3: Forecasting & anomaly logic

Phase 4: Frontend UI with Streamlit

Phase 5: Pinecone integration

Phase 6: Voice & report modules

6. Functional and Performance Testing

- Forecasting & anomaly detection validated with sample data
- Chatbot tested for urban queries

- File uploads and dashboards responsive
- Performance Benchmarks:**

- Chat API: <500ms
- Forecast: <2s
- News Summary: <3s

7. Results

7.1 Output Screenshots

Figure 1: Chat Assistant

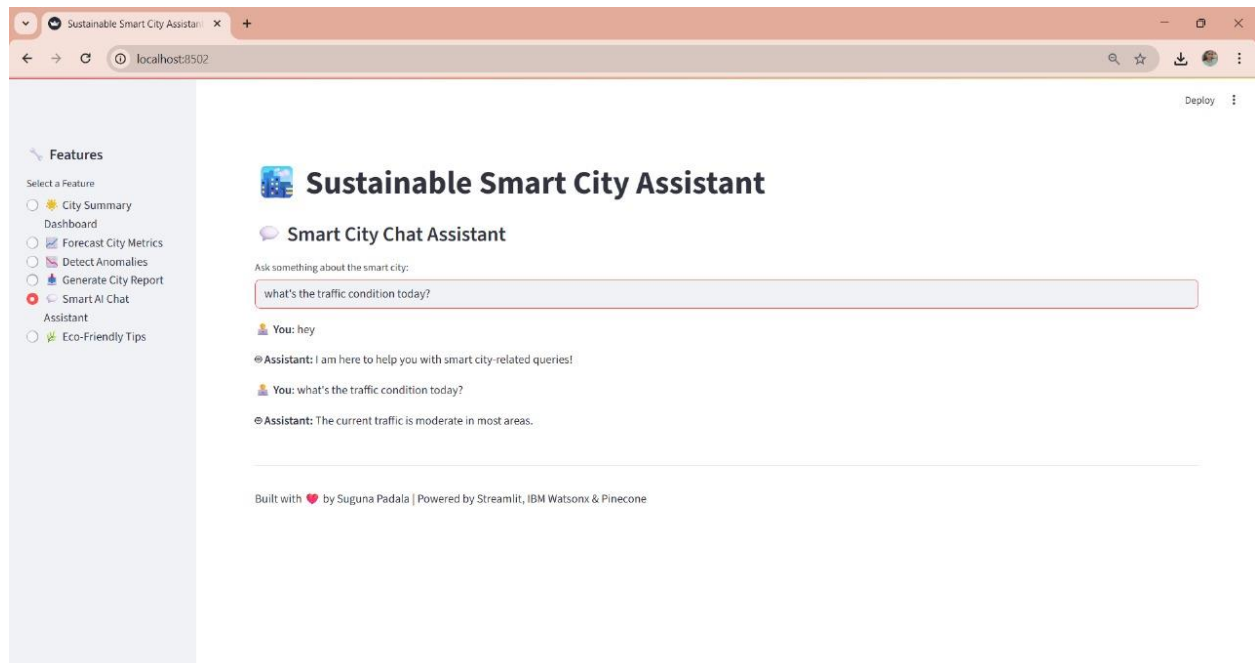


Figure 2: Report Generation

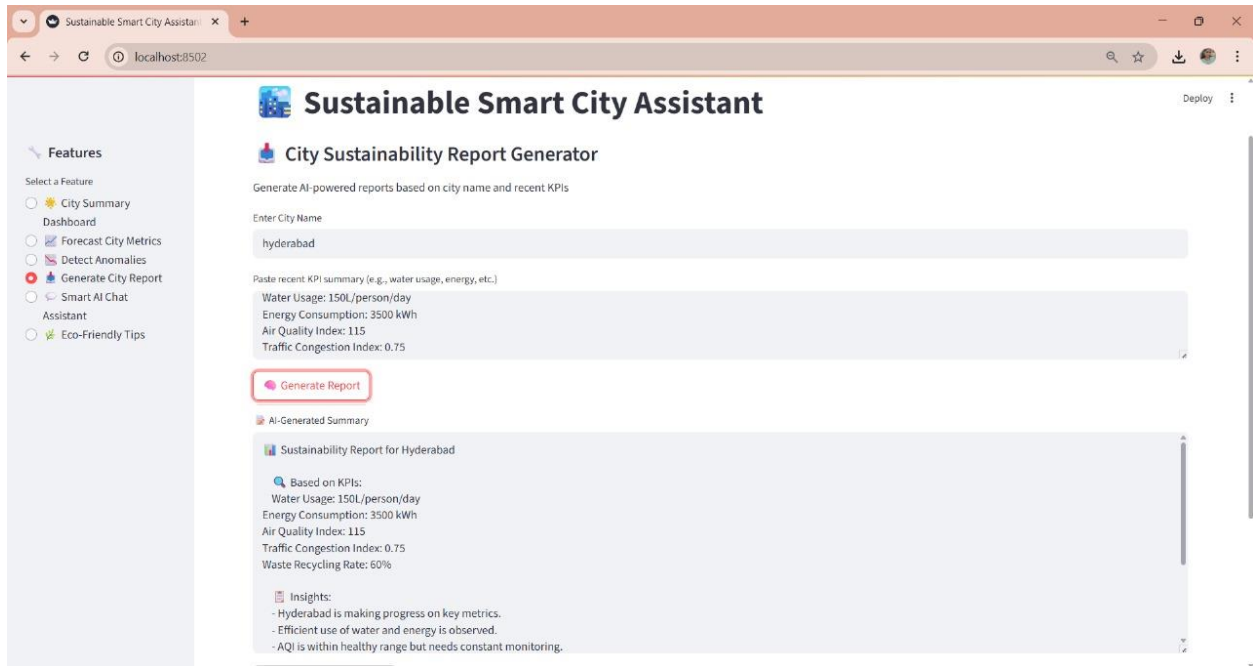


Figure 3: CSV Upload & Forecast

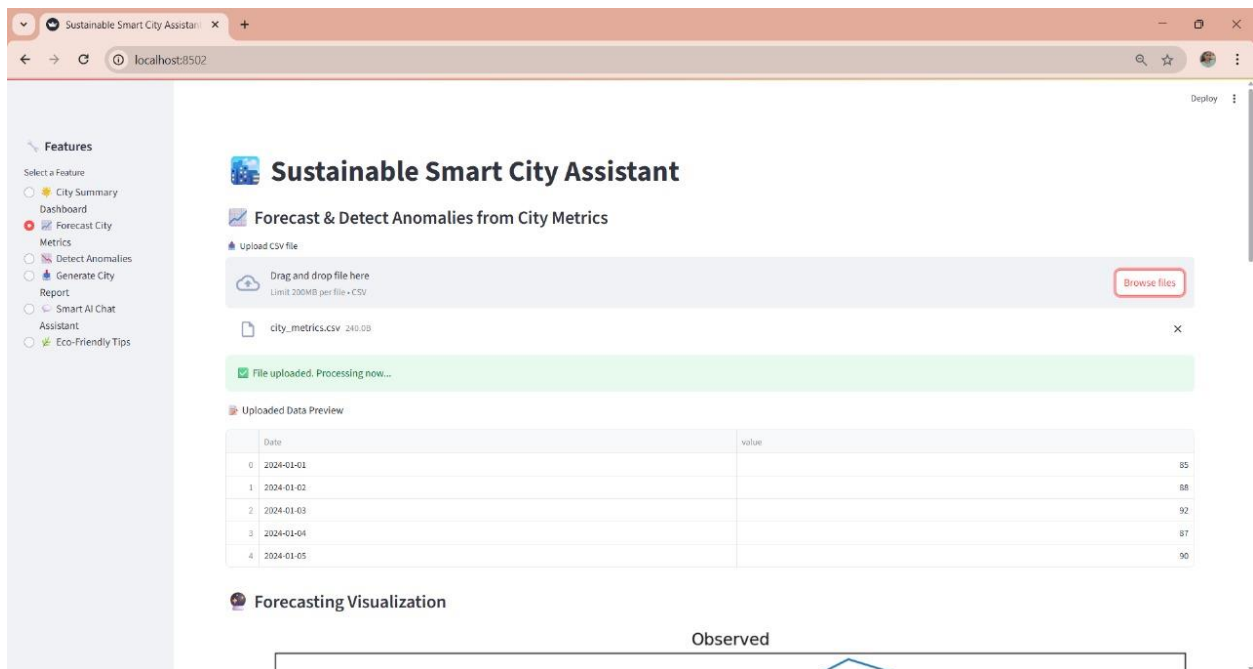


Figure 4: Anomaly Detection Graph

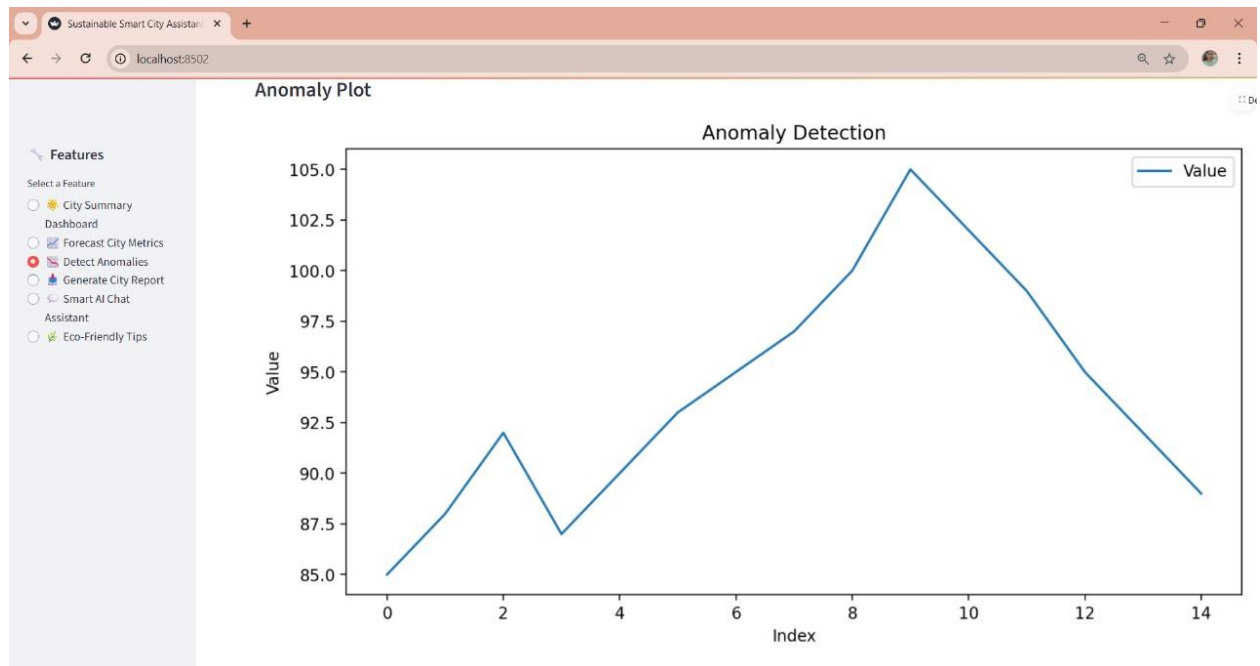
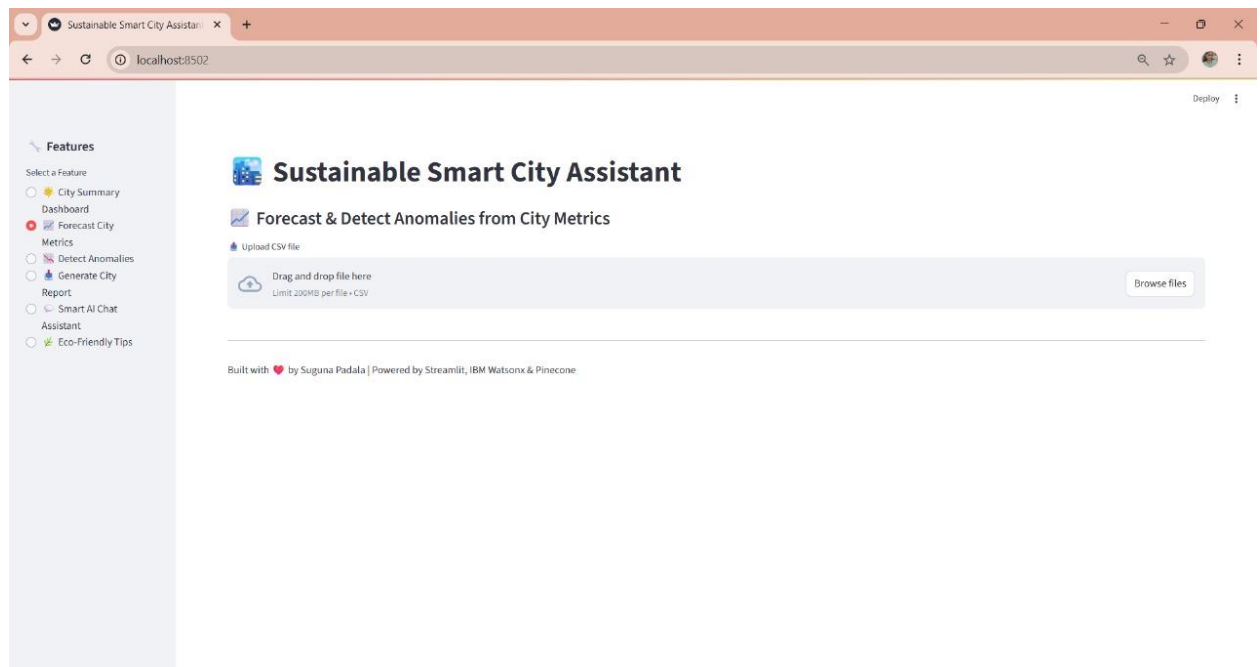
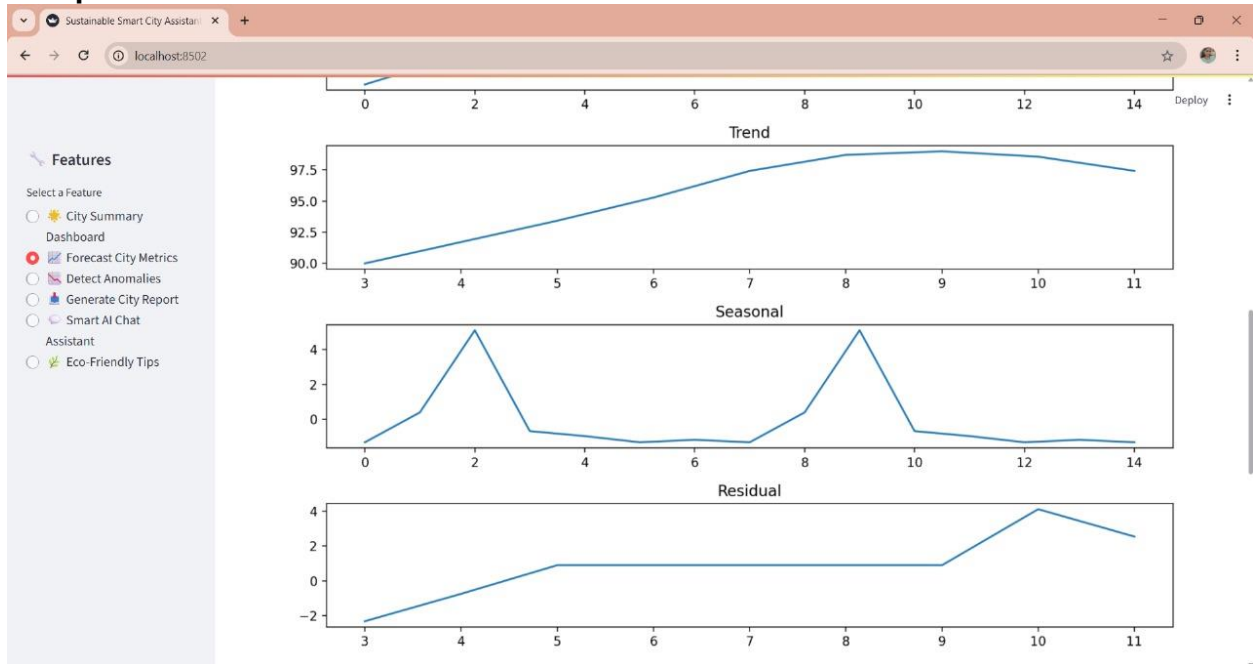


Figure 5: KPI Dashboard



7.2 Sample Outputs

Sample Forecast:



8. Advantages & Disadvantages

8.1 Advantages

- User-friendly Streamlit UI
- Modular FastAPI backend
- Real-time assistance with IBM LLM
- Extensible with future integrations

8.2 Disadvantages

- Relies on CSV, no DB by default
- No user auth or multi-role setup
- Voice/semantic features not fully implemented

9. Conclusion

The Smart City Assistant demonstrates a viable approach to sustainable city analytics using LLMs, forecasting models, and real-time tools. It meets current planning needs while offering a roadmap for scalable innovation.

10. Future Scope

- Role-based authentication (OAuth2/JWT)
- DB storage (SQLite/MongoDB)
- Real-time IoT integration
- Map route planning APIs (Google/TomTom)
- Multilingual support & voice assistant
- Domain-specific AI news curation

11. Appendix

11.1 GitHub & Project Demo Link

[Project Demo Video URL](#)

[My Github Repo](#)